POWER SUPPLY SWITCHING

id092100205000

Outline

- The power supply circuit can be switched by pressing the bush button start.

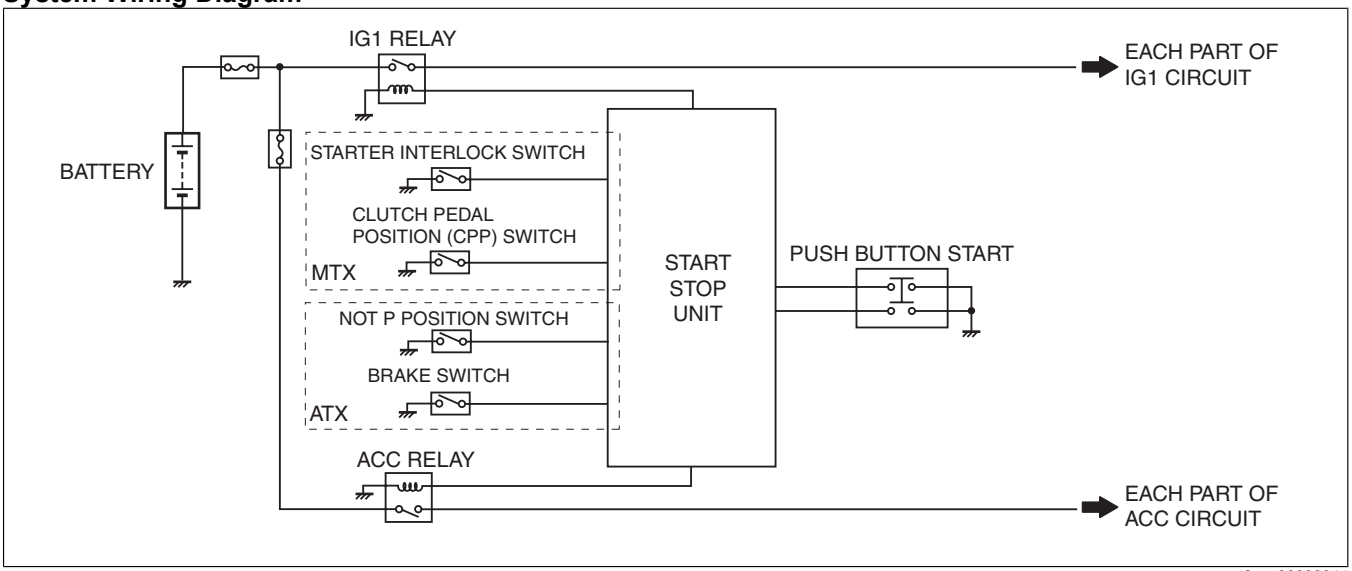
Function

- Based on the push button start signals, the start stop unit switches the power supply circuit by turning on/off the relay according to the vehicle conditions such as the position of the selector lever (ATX)/shift lever (MTX) and depression/release of the brake pedal (ATX)/clutch pedal (MTX).

Automatic ignition switch off function

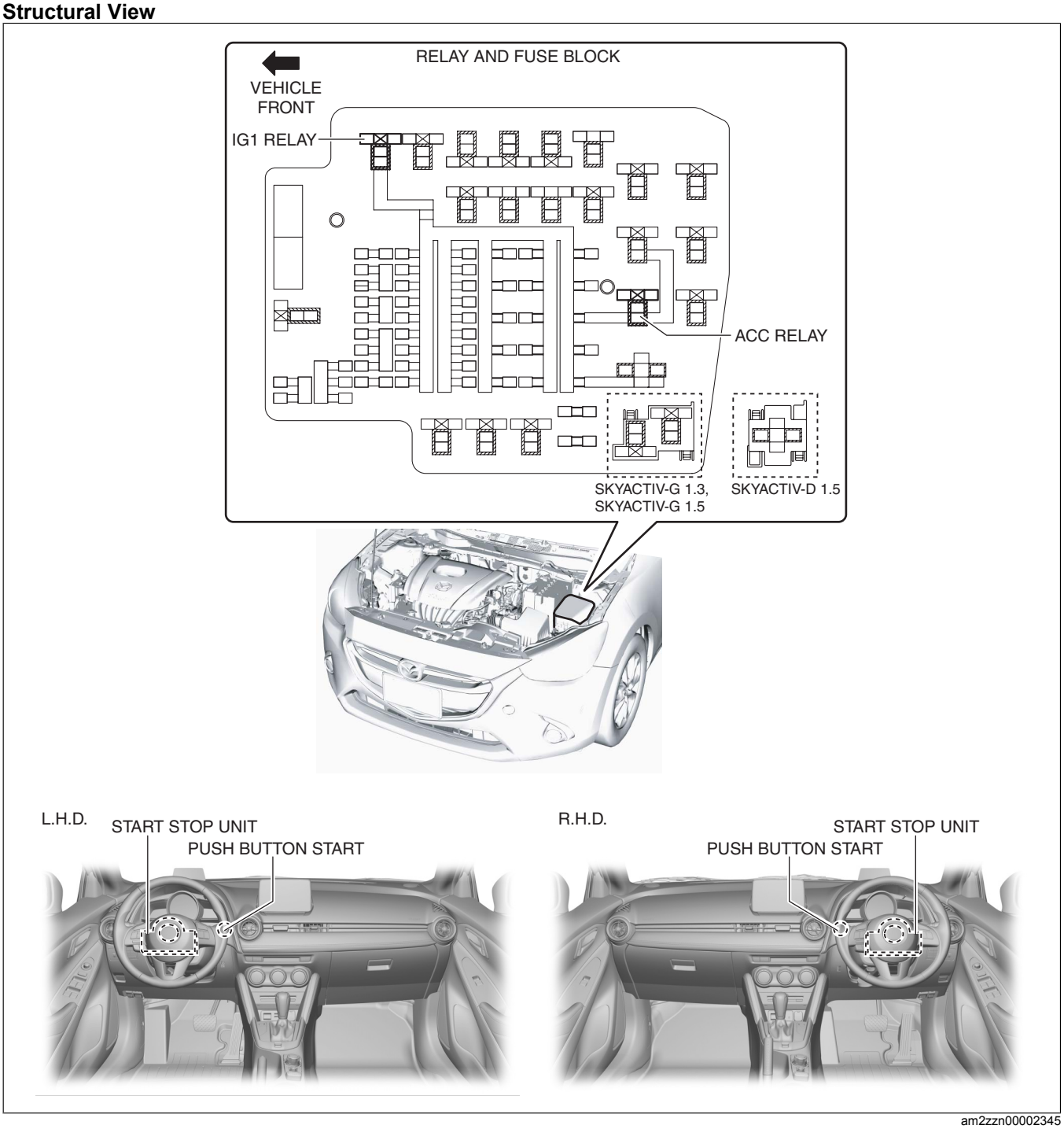
- When 25 min or more have elapsed with the ignition switched to ACC, the ignition is automatically switched off.

System Wiring Diagram



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Structural View



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Operation

Switching the ignition

- By pressing the push button start under the conditions shown in the figure, the ignition can be switched.

MTX

ENGINE SWITCH STATUS	ENGINE STATUS	CLUTCH PEDAL STATUS	
		NOT DEPRESSED	DEPRESSED
OFF *1	OFF		
ACC *1	OFF		
ON	OFF		
START	CRANKING		
ON	RUNNING	*2	*2

○ : PRESS PUSH BUTTON START

↓ : IGNITION SWITCH TRANSITION

ATX

ENGINE SWITCH STATUS	ENGINE STATUS	BRAKE PEDAL STATUS					
		NOT DEPRESSED			DEPRESSED		
		SELECTOR LEVER POSITION					
		P POSITION	N POSITION	OTHER THAN P, N POSITION	P POSITION	N POSITION	OTHER THAN P, N POSITION
OFF *1	OFF						
ACC *1	OFF						
ON	OFF						
START	CRANKING						
ON	RUNNING	*2			*2	*2	

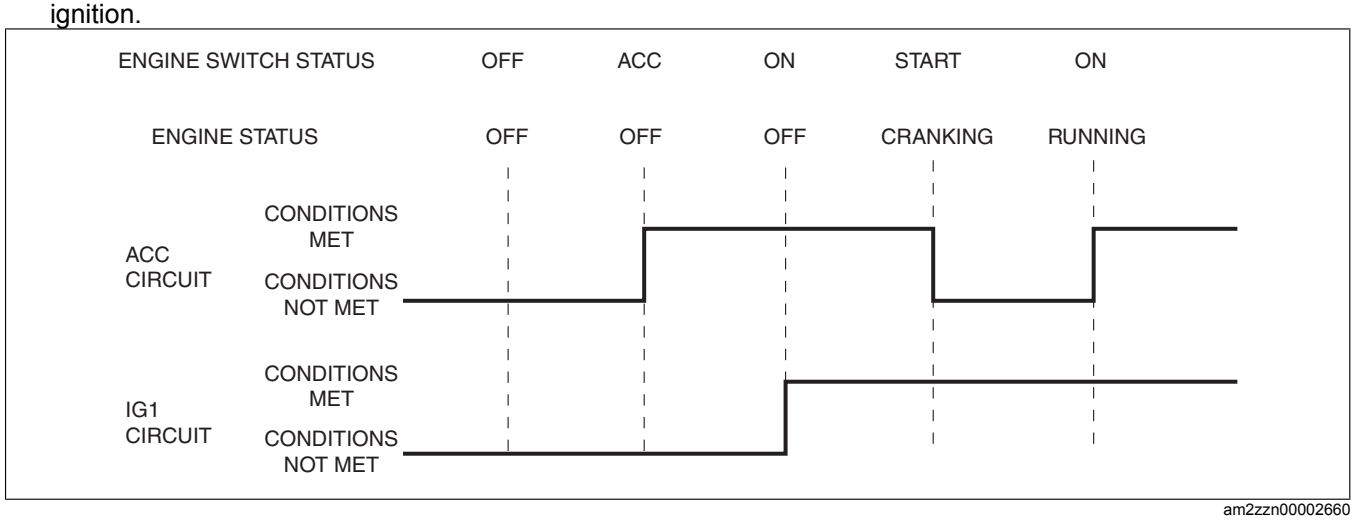
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*1 : To switch the ignition to ACC or OFF from ON (engine on), the vehicle speed must be 5 km/h {3 mph} or less.

*2 : To improve engine re-startability, it may take time to switch the ignition off. Refer to the [VARIABLE VALVE TIMING MECHANISM] for details. (See VARIABLE VALVE TIMING MECHANISM [SKYACTIV-G 1.3, SKYACTIV-G 1.5 (WITH 4-1 EXHAUST SYSTEM)].)

Switching the power supply circuit

- The start stop unit switches the power supply circuit as shown in the table according to the switching of the ignition.



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Automatic ignition switch off function

- When all the following conditions are met for **25 min or more**, the start stop unit switches the ignition off automatically.

— ACC relay on and IG1 relay off (ignition switch ACC condition) are detected.
— All door latch switches and liftgate latch switch off (all doors and liftgate are closed) is received
— Push button start, light switch, wiper and washer switch, steering switch, and hazard warning switch are not operated.
— Vehicle speed of **5 km/h or less** is received
— Clutch pedal position switch and starter interlock switch off signal (clutch pedal is not depressed) is detected (MTX)
— Brake switch off signal (brake pedal is not depressed) is detected (ATX)
— NOT P position switch off (selector lever is in P position) is detected (ATX)

Fail-safe

- Not applicable